

# CODE SPORT



In this month's column, we feature a set of interview questions.

I was asked by some of our readers to put together interview questions for systems software engineers to practice on. Instead of focusing on specific areas like algorithms or computer architecture, I have put together a medley of questions from different areas, which can be used for mock interviews. In a real interview, the questions tend to be typically more open-ended rather than have a fixed correct answer. The typical expectation of the interviewer is to see whether the candidate can think broadly while working through constraints and also propose multiple alternatives/solutions with different cost-benefit tradeoffs. So there may not be a single fixed answer to some of the following questions.

1. What are the resource control mechanisms available in Linux? Are you familiar with Cgroups?
2. You are managing two different applications that need to run on the same server. Application 1 is CPU-hungry most of the time whereas Application 2 is mostly disk bound, but has short bursts of time when it gets CPU-hungry. Application 1 also has periods when it has user interactions. If you want to ensure reasonable performance to both applications, without any visible latency in the user interactions of Application 1, what kind of resource control mechanism would you come up with?
3. What are the Linux file systems you are familiar with? While file systems are typically part of the Linux kernel, FUSE allows file systems to be implemented in user space. If you are asked to implement a FUSE file system in user space for a specific purpose like encryption, can you explain your approach?
4. You have developed an application that consumes various system resources like the CPU, memory, disk and network. You are asked to profile the

application to identify and break down where it is spending its time and how it utilises the resources. What tool would you use? Explain your profiling methodology in detail.

5. Are you familiar with the tools SAR (system activity reporter) and Collectl? Have you used them?
6. What are erasure codes? How do they improve the reliability of data storage systems?
7. What are object-based storage systems? Can you name any of them? How do they differ in behaviour from traditional file systems?
8. You have an application in which you need to encode sensitive information such as the user's credentials in a string and store it in a variable in the application instead of obtaining it dynamically at runtime. You want to avoid malicious users reading this string inadvertently. If you store the string in a character array, by using the 'strings' command, anyone can get the string and can use it. Another possibility is to encode the string using some known convention and decode it at the point of use. By disassembling the executable and reverse engineering, a malicious user can still get the sensitive information. What are the ways in which you could address this issue?
9. As part of a large application, you have the need for a sorting routine. You have on hand two sort routines, one based on quick sort and another on merge sort. Which sorting technique would you choose, and what criteria would your selection be based on?
10. Do you know the difference between statically compiled languages and dynamically compiled languages? Is Java a statically compiled language?
11. Your company has a legacy application that